AUDITORY GROUPING DAILY

Sunday, 12 July 2026 · 295 papers screened · curated

Today's batch skews toward auditory circuit mechanism and predictive coding rather than sequence-behavior tasks, but it holds together: two papers dissect how interneuron types and cortical layers implement scene segregation and prediction-error signaling, one shows a primate frontal region categorically encoding vocalizations with your own recording toolkit, and one is a stimulus-design cross-check for your mouse work's hearing-loss-constrained frequency range. Nothing today bears on evidence-accumulation or GLM strategy modelling, so that section is empty rather than padded. The clearest tie to your work is the parvalbumin/somatostatin dissociation in auditory-cortex scene analysis, a circuit-level precedent for how interneuron subtypes could shape target-versus-distractor representations before a licking decision. Today's canon pick, Saffran, Aslin & Newport (1996), anchors this theme as the founding demonstration that sequence segmentation can run on transitional-probability tracking alone.

CHUNKING & SEQUENCE PROCESSING

Parvalbumin and somatostatin interneurons support distinct computations during auditory scene analysis

BIORXIV *Qu, Z. et al. — bioRxiv — 2026-07-10*

Corresponding: Howard Gritton — University of Illinois Urbana-Champaign — United States | Open paper

Using a cocktail-party-style paradigm with cell-type-specific optogenetic suppression in awake mouse auditory cortex, this preprint dissociates two interneuron classes: parvalbumin (PV) neurons broadly reshape spatial receptive-field structure (tuning width, centroid, sparseness) and support target discriminability even without any competing sound, whereas somatostatin (SST) neurons leave spatial tuning largely intact but are selectively required for discriminating a target once a spatially separated masking sound is present. This is a direct circuit-level precedent for how inhibitory interneuron subtypes in auditory cortex could differentially shape the representation of your trained target sequence versus a distractor sequence before the signal is available for a licking decision, and it's worth asking whether an analogous split exists for your temporal (rather than spatial) form of scene segregation once you extend recordings from prelimbic cortex into auditory cortex.

Auditory Streaming and Interoception

OPENALEX *John Veillette et al. — UC San Diego — 2026-07-10*

Corresponding: Howard Nusbaum — University of Chicago — United States | Open paper

This is a newly deposited dataset (not a findings paper) combining behavioural auditory-streaming judgments with electrophysiological recordings of interoceptive signals, aimed at testing how internal bodily state influences auditory perceptual grouping. It's a resource worth knowing about rather than a result to cite, but the framing question, whether an internal physiological state variable modulates how a listener parses a sound stream into objects, is a useful analogy for your own GLM Module C, which tries to capture slow internal-state drift (satiety, time-on-task) shaping sequence-discrimination performance session to session.

AUDITORY & PREFRONTAL ELECTROPHYSIOLOGY

Selective Neural Responses to Conspecific Vocalizations in Marmoset Area 32.

PUBMED *Kevin D Johnston et al. — Journal of Neuroscience — 2026*

Corresponding: Kevin D Johnston — University of Western Ontario — Canada | Open paper

Recording single units with Neuropixels and Utah arrays in marmoset pregenual anterior cingulate cortex (area 32) during playback of marmoset, macaque, human, and non-vocal sounds, this study found that 21% of 1,713 recorded neurons were categorically selective, with the strongest and earliest responses (decoding above chance within 50-100 ms) to conspecific calls, and that representational similarity analysis separated marmoset calls into their own cluster distinct from other sound categories. This directly parallels your human work's core question of how frontal cortex builds categorical representations out of continuous acoustic input, and the combination of high-density single-unit recording, population decoding, and representational similarity analysis is essentially the same analysis toolkit used in your lab's prior single-neuron finding that prefrontal cortex carries semantic rather than purely acoustic/phonological structure.

STATISTICAL LEARNING & ODDBALL

Temporal dynamics of contextual processing in the inferior colliculus of a rat model of autism: Sex- and age-dependent trajectories

CROSSREF (MODEL-DISCOVERED) *S. Cacciato-Salcedo et al. — Hearing Research — 2026-07*

Corresponding: Manuel S. Malmierca — University of Salamanca — Spain | Open paper

Recording in the inferior colliculus of rats, this study from Malmierca's group found that normal maturation reorganizes prediction-error signalling and repetition suppression in a sex-specific way, with females developing longer response latencies and stronger prediction-driven responses, while prenatal exposure to a chemical autism model selectively disrupted this developmental trajectory in females. It's a reminder that the prediction-error/oddball framework underlying your own interest in deviance detection and regularity extraction is already known to vary by sex and developmental stage at the subcortical stage of the auditory pathway, upstream of the cortical regions you record from, which is worth keeping in mind when interpreting session-to-session or animal-to-animal variability in your own auditory-cortex data.

When vision listens: auditory predictive processing in primary visual cortex of anesthetized rats

CROSSREF (MODEL-DISCOVERED) *Jazmín S. Sánchez et al. — Hearing Research — 2026-09*

Corresponding: Manuel S. Malmierca — University of Salamanca — Spain | Open paper

Also from Malmierca's group, this study recorded primary visual cortex in anesthetized rats during auditory oddball sequences and found that infragranular layers V-VI carry genuine prediction-error signals (stronger to deviant than standard tones, amplified under temporally uncertain conditions) rather than simple repetition suppression, with layer V faster/phasic and layer VI broader/sustained. It extends the predictive-coding, deviance-detection framework you draw on for your own oddball/regularity-extraction interest outside the auditory pathway entirely, which is useful context for how domain-general (versus auditory-specific) you should expect any prediction-error signal you find in prelimbic or auditory cortex to be.

LANGUAGE, STROKE & BCI

Dynamic acoustic-to-categorical representations of phonemes and prosody along ventral and dorsal speech streams.

PUBMED *Seung-Cheol Baek et al. — Nature Communications — 2026*

Corresponding: Daniela Sammler — Max Planck Institute for Empirical Aesthetics — Germany | Open paper

Combining MEG with representational similarity and multivariate transfer-entropy analyses, this study found that acoustic and categorical representations of phonemes and prosody form in parallel (not serially, as traditional dual-stream models predict) along both the ventral and dorsal speech pathways, with prosody categories extending further along both streams than phoneme categories. This is a direct precedent for your human chunk-in-stream task's question of where and when a bottom-up acoustic signal becomes a top-down template-matched category, and the parallel-rather-than-serial finding is worth checking against your own high-gamma signal across the stimulus classes that extend from chords toward speech (synthetic vowels, open syllables).

CIRCUITS, METHODS & POPULATION ANALYSIS

Molecularly defined auditory neuron subtypes show different vulnerabilities to noise- and age-related synaptopathy in mice.

PUBMED *Joy A Franco et al. — Nature Communications — 2026*

Corresponding: Lisa V. Goodrich — Harvard Medical School — United States | Open paper

Using genetic labelling in mice, this study found that of the three spiral ganglion neuron subtypes carrying signal from the cochlea to the brain, one subtype (Ia) has larger post-synaptic densities and is resilient to both noise- and age-related synapse loss, while the other two (Ib/Ic) are vulnerable to noise-induced (but not age-related) synaptopathy, and that genetically reprogramming Ib/Ic neurons toward an Ia-like identity is protective. This bears directly on your mouse task's stimulus-design rationale, since your whole chord-frequency range (3-19 kHz) was chosen specifically to survive C57BL/6 progressive high-frequency hearing loss; this paper is a reminder that even within a frequency range chosen to avoid outright hearing loss, differential spiral-ganglion-neuron-subtype vulnerability to repeated sound exposure could still be a slow confound across the many chronic-recording sessions your task requires.

FOUNDATIONS PICK — read this classic

Saffran, Aslin & Newport (1996, Science) — statistical learning in infants.

Foundations pick — abstract only

- Using intracranial EEG recorded directly from cortex in neurosurgical patients, and stimuli built to have naturalistic statistical regularities rather than the simple repeating patterns most prediction studies use, the authors report that time-forward prediction signals were carried mainly by low-frequency activity spread across sensory, parietal, and frontal cortex, while prediction-error signals showed up in both low- and high-frequency activity, with the high-frequency component largely confined to sensory regions.
- The paper's central claim is a negative result for a common assumption: prediction and prediction-error signals did not show the clean spatial or spectral segregation that earlier hypotheses expected, meaning you can't assume a given frequency band or region is doing only one of those two jobs at once.
- Directed connectivity between regions dropped over the course of a sequence as its predictability increased, except along pathways originating in parietal cortex, which stayed elevated — the abstract frames this as evidence for an integrative, distributed predictive process with a sustained top-down contribution from parietal cortex and the dorsal auditory pathway, rather than prediction being resolved once and then handed off.
- Because the abstract itself doesn't specify exactly what 'naturalistic statistical regularities' means for the tone sequences (only that they're closer to natural auditory statistics than typical lab stimuli), this is the one claim in the paper worth verifying against the full Methods before you lean on it too hard as a direct analogue to your own tone/chord sequences — the abstract's framing is credible but the specific regularity structure isn't described at this level of detail.

HOW TO READ IT (≈20 min)

Full text wasn't available for this pass (Elicit access is also not currently enabled on this account), so treat this as an abstract-only read plan to refine once you have the PDF. Start with the Introduction to see their specific critique of the low-frequency-predicts/high-frequency-errors assumption (2-3 min). Go straight to Methods and read closely exactly how the 'naturalistic' tone-sequence statistics were generated and which electrode coverage regions counted as sensory/parietal/frontal (5-6 min) — this is the part the abstract underspecifies and where your own judgment about relevance to tone/chord sequences will actually be decided. Read the Results on spatial/spectral overlap in full (6-7 min). Read the directed-connectivity results carefully, especially the parietal-cortex exception (4-5 min). Skim the Discussion for how they reconcile the overlap finding with predictive-coding theory (2 min).

YOUR QUESTION TO ANSWER

The abstract reports that parietal-cortex outgoing connectivity stayed high even as overall directed connectivity fell with increasing predictability — if that finding survives a look at the full Methods, would you expect an equivalent 'exception' region in your own mouse prelimbic-versus-auditory-cortex comparison (i.e., a region whose outgoing influence on the other stays elevated even once the mouse has learned the target sequence well), or does your task's much smaller state space make that kind of sustained top-down channel less likely to be needed? And is your current pipeline (single-unit and LFP recordings in prelimbic cortex, later auditory cortex) actually set up to compute directed connectivity between the two regions, or would testing this require adding a specific analysis you haven't run yet?

Source status — Flagship journals (Tier 1): 234 papers | bioRxiv deep scan (Tier 2): 1 papers | OpenAlex (Tier 3): 163 papers | PubMed (Tier 3): 3 papers | arXiv (Tier 3): 0 papers